

ABSTRACT

DEEP LEARNING APPROACHES FOR CTENOPHORE IDENTIFICATION AND TRACKING

by
Anagha Bharadwaj

Ctenophores are translucent marine organisms with nearly invisible tentacles and pose significant challenges due to their transparent morphology and ambiguous structural features. This research addresses the classification and tracking of these organisms and its performance under sparse-data environments.

A dataset from the NJIT Life History Lab consisting of microscopic laboratory videos and photographs of different growth stages is used to train and assess a number of convolutional neural network designs, including VGG16, ResNet, BioCLIP2, YOLO, and DeepLabCut. Additionally, a web-based interface is developed to evaluate expert-labeled ground truth with the model's performance.

The findings demonstrate that existing computer vision frameworks are in moderate agreement with expert assessments which is inadequate for dependable automated analysis of ctenophores. Improvement in performance and accuracy requires larger datasets along with models specifically designed for translucent marine creatures.

My lawyer advises me to include this. ¹

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by
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APPROVAL PAGE

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IDENTIFICATION AND TRACKING**

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